



Liquid-liquid equilibria for azeotropic mixture of methyl tert-butyl ether and methanol with ionic liquids at different temperatures



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ABSTRACT

Searching for green solvents is imperative for the sustainable development of liquid-liquid extraction. Liquid-liquid equilibrium data for two ternary systems of methyl tert-butyl ether (MTBE) + methanol + N-methylimidazolium hydrosulfate ([MIM][HSO₄]) and MTBE + methanol + 1-butyl-3-methylimidazolium hydrosulfate ([BMIM][HSO₄]) were measured at 278.15 K, 298.15 K and 318.15 K at 101.325 kPa for this study. The values of the distribution coefficient and the separation factor were calculated based on the experimental data, and then, the extraction ability was determined. Factors affecting the phase behavior such as the nature of the ionic liquid and the temperature were discussed. NRTL and UNIQUAC models were applied to correlate the experimental tie-line data.

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1. Introduction

MTBE was proposed as “the third generation of petrochemicals” in the 1980s, and it is one of a group of chemicals commonly known as “oxygenates”. MTBE can significantly improve the octane number of gasoline [1–4] and has many advantages such as being colorless and transparent and having low viscosity and good anti-knocking properties [5,6]. At the same time, MTBE can be cleaved into isoprene to form rubber and other chemical raw materials [7], and the highest-quality MTBE is a pharmaceutical intermediate.

MTBE can be obtained from the exothermic reaction from mixing butene and methanol in the presence of an acidic catalyst [8], and a mixture containing MTBE and feedstock material methanol is produced in this process. Because of the existence of an azeotrope incorporating MTBE and methanol, conventional distillation cannot effectively implement the separation for obtaining a high-purity MTBE product. Although special distillation technologies such as extractive distillation [9,10], azeotropic distillation [4], and press-swing distillation [11] are commonly used to separate azeotropes for achieving target separations, these technologies have high energy consumption. Liquid-liquid extraction [12–14] takes advantage of the solubility differences of components in

solvents to achieve mixture separation and is an environmentally friendly technique widely used to separate azeotropes in chemical industry. The energy consumption of liquid-liquid extraction is generally lower than the energy consumption of the distillation process [15].

The determination of extraction agent is vital for the liquid-liquid extraction process and can influence the separation efficiency. The traditional organic solvents are highly volatile and toxic, leading to negative environmental effects. Ionic liquids (ILs), as new extraction agents, are called “designer solvents”, and their structures can be altered to adjusted physical and chemical characteristics for specific applications [16,17]. ILs are widely used in many fields due to their designability and distinct advantages for physical and chemical properties [18–22]. Specifically, ILs show outstanding advantages in liquid-liquid extraction as new extracting agents.

The recovery of extractant has important significance in industrial applications. The recovery methods of commonly used extractants include distillation and flashing [23]. According to the boiling point of the organic solvent and the ILs, distillation is suitable for organic solvents with lower boiling points, and flashing is suitable for ILs with higher boiling points. The equipment cost for flashing is lower than the equipment cost for distillation. Therefore, from an economic perspective, ILs are also great alternative solvents.

Monika et al. investigated the extraction for separating hexane/hex-1-ene and cyclohexane/cyclohexene using dicyanamide-based

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ILs as extraction agents and found that (3-cyanopropyl) methylpyrrolidinium dicyanamide and N-ethyl-N-methylmorpholinium dicyanamide exhibit the best separation performance for hexane/hex-1-ene and cyclohexane/cyclohexene, respectively [24,25]. Cai et al. studied different phosphoric-based ILs for separating the azeotrope of ethanol and hexane, and 1,3-dimethylimidazolium dimethylphosphate showed the highest selectivity [26–28]. Morteza et al. measured LLE properties for the extractive desulfurization with 1-butyl-3-methyl imidazolium thiocyanate and correlated the experimental data using the NRTL model [29]. Amparo et al. studied the LLE data for the systems of water + ethanol + ILs (1-butyl-1-methylpyrrolidinium bis(trifluoromethylsulfonyl)imide [BMP][NTf₂] and 1-butyl-3-methylimidazolium hexafluorophosphate [BMIM][PF₆]) at different temperatures [30,31]. Lesly et al. investigated the LLE data on an alcohol solution with the new IL tetraoctyl ammonium 2-methyl-1-naphthoate, and a good separation of an alcohol and water azeotrope was obtained [32–34].

The LLE data for two systems of MTBE + methanol + [BMIM][HSO₄] and MTBE + methanol + [MIM][HSO₄] at 278.15 K, 298.15 K and 318.15 K at 101.325 kPa were measured in this work. The NRTL and UNIQUAC activity coefficient models [35,36] were used to correlate the experimental data for the two systems. The distribution coefficient (β) and separation factor (S) were calculated based on the experimental LLE data, and the feasibility of the ILs in the liquid-liquid extraction was evaluated.

2. Experimental

2.1. Chemicals

The purity of ethanol, methanol and MTBE was evaluated using gas chromatography (SHIMADZU, GC-2014 C). The purity of the ILs [BMIM][HSO₄] and [MIM][HSO₄] were no less than 98.0%. The detailed information including source, CAS number, molar mass (M) and purity are presented in Table 1. The IL structures are presented in Fig. 1. All the materials in the experiment were used without further purification.

2.2. Experiment and procedure

The LLE data for the two systems of MTBE (1) + methanol (2) + [BMIM][HSO₄] (3) and MTBE (1) + methanol (2) + [MIM][HSO₄] (3) were collected at 278.15 K, 298.15 K and 318.15 K at 101.325 kPa. The experimental procedure and equipment used for measuring the LLE were depicted in detail in our previous works [37–39]. The masses of MTBE, methanol and ILs were determined by using FA-1204B electronic scales (made by Shanghai Tianmei Balance Instrument Co., Ltd.). Then, the mixtures of MTBE (1) + methanol (2) + [BMIM][HSO₄] (3) and MTBE (1) + methanol (2) + [MIM][HSO₄] (3) were placed into a self-designed equilibrium cell [38], and a stir bar was used to stir the mixtures vigorously.

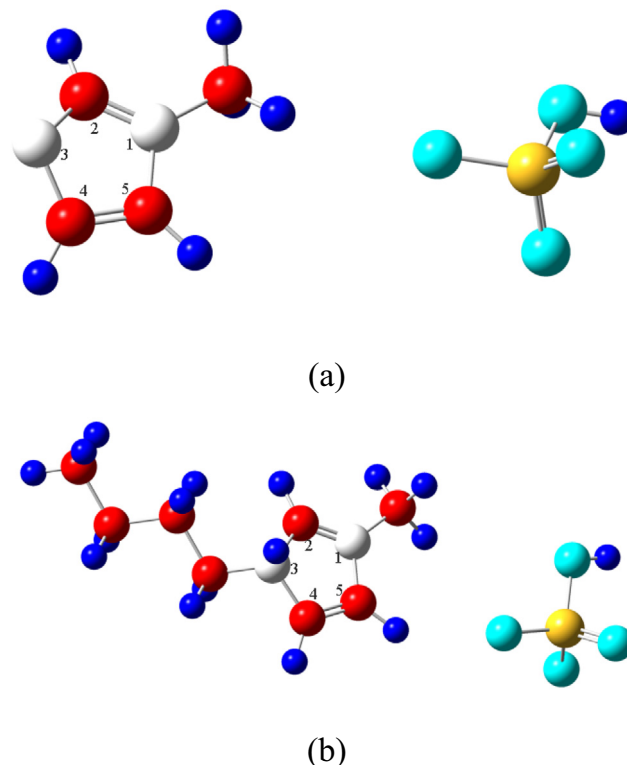


Fig. 1. Ionic liquids structures; (a) [MIM][HSO₄]; (b) [BMIM][HSO₄].

After 3 h of intense stirring (approximately 1000 r/min) in the equilibrium cell, the mixture was heated for 15 h to ensure that the two liquid phases (MTBE-rich phase and IL-rich phase) separated completely. In the above process, the samples should be maintained at a constant temperature by connecting the jacket of the equilibrium cell to a HX-105 low-temperature thermostat, for which the temperature fluctuation is 0.05 K.

After the two phases separated completely, the IL-rich phase and MTBE-rich phase were removed from the LLE equipment, and the internal standard (ethanol) was added into each sample. Each sample prepared was analyzed by GC with a Porapak Q packed column (3 m × 0.04 m) and a thermal conductivity detector (TCD). Because the ILs are nonvolatile and the accumulation of ILs will reduce the service life of columns, the samples pass through a precolumn before entering the chromatographic column. The carrier gas used in the experiment is high-purity helium gas (>99.99%). When testing the sample, the temperature of the injector, detector and oven were 473.15 K, 473.15 K and 503.15 K, respectively. Under these conditions, these components could be well separated with preferable peak shapes and completely separated peaks. The retention times of methanol, ethanol and MTBE

Table 1

Sources, CAS number, molar mass, purification, water contents by mass, w_w , and analysis method of the chemicals used in this work.

Chemical	Source	CAS number	Molar mass/(g·mol ⁻¹)	Mass purity stated by supplier	$w_w/10^{-6}$	Analysis method
MTBE	Hushi Sinopharm Group Chemical Reagent Co., Ltd.	1634-04-4	88.1482	0.990 ^a		GC
Ethanol	Tianjin Kermel Chemical Reagent Co., Ltd.	64-17-5	46.07	0.995 ^a		GC
Methanol	Tianjin Kermel Chemical Reagent Co., Ltd.	67-56-1	32.04	0.998 ^a		GC
[BMIM][HSO ₄]	Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences.	262297-13-2	236.29	0.980 ^a	<500 ^a	
[MIM][HSO ₄]	Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences.	681281-87-8	180.18	0.980 ^a	<500 ^a	

^a Analysis by supplier.

were approximately 2.6 min, 4.2 min and 12.8 min, respectively. Each prepared sample was measured three times at minimum. GC cannot be used to analyze the composition of the ILs because of the nonvolatility of the ILs, and the composition of the ILs was analyzed using a mass difference method. The samples (MTBE and methanol) were evaporated in a vacuum drying oven for 48 h at a fixed temperature of 373.15 K. Each sample was accurately weighed with electronic scales before and after evaporation.

3. Results and discussion

3.1. Experimental data

The LLE data for two ternary systems of MTBE (1) + methanol (2) + [BMIM][HSO₄] (3) and MTBE (1) + methanol (2) + [MIM][HSO₄] (3) were measured at 278.15 K, 298.15 K and 318.15 K at 101.325 kPa, and the experimental results of w_{if} , w_i , β and S are presented in Table 2, in which w_{if} and w_i represent the mass fraction of the feed composition and the mass fraction of the component in every sample in this work, respectively. Uncertainties of the experimental LLE data on mass fraction w_i , distribution coefficient β and separation factor S for ternary systems of MTBE (1) + methanol (2) + ILs (3) were calculated according to the method from the literature [41–42] and are presented in Table S1. Figs. 2 and 3 illustrate the experimental results for the six LLE systems (MTBE + methanol + ILs). The phase behaviors of all the studied systems show type I. Methanol is completely miscible in MTBE, [BMIM][HSO₄] and [MIM][HSO₄], while MTBE is partially miscible in [BMIM][HSO₄] and [MIM][HSO₄]. The immiscibility regions of the two systems of MTBE + methanol + [BMIM][HSO₄] and MTBE + methanol + [MIM][HSO₄] are sufficiently large for liquid-liquid extraction.

β and S calculated based on the experimental data were used to evaluate the extractive performance of the ILs for the purity of the methanol from MTBE. β and S can be calculated by the following equations:

$$\beta = \frac{w_2^II}{w_2^I} \quad (1)$$

$$S = \frac{(w_2^II/w_1^II)}{(w_2^I/w_1^I)} \quad (2)$$

where w_2^I and w_2^{II} present the mass fraction of methanol in the MTBE-rich phase and IL-rich phase, respectively; w_1^I and w_1^{II} present the mass fraction of MTBE-phase in the MTBE-phase and IL-rich phase, respectively. The calculated values of β and S for the studied systems at 278.15 K, 298.15 K and 318.15 K are listed in Table 2, and the graphical representations are shown in Figs. 4–5, respectively. The value of S is greater than unity, which indicates that [BMIM][HSO₄] and [MIM][HSO₄] are suitable solvents for extracting methanol from the MTBE-methanol mixture. With the increase in w_2^I , the β of [BMIM][HSO₄] increases when $w_2^I < 0.10$ and decreases when $w_2^I > 0.10$ (Fig. 4). With the increase in w_2^I , the S of [MIM][HSO₄] decreases rapidly when $w_2^I < 0.20$ and decreases slowly when $w_2^I > 0.20$ at each temperature (Fig. 5). The β and S of [MIM][HSO₄] decrease with increasing methanol concentration in the IL-rich phase at each temperature. The values of β and S have a slight increase from 278.15 K to 298.15 K and have an obvious increase from 298.15 K to 318.15 K for the two systems of MTBE + methanol + [MIM][HSO₄] and MTBE + methanol + [BMIM][HSO₄], indicating that the extraction capacity of [BMIM][HSO₄] and [MIM][HSO₄] increases with increasing temperature.

Many researchers explored the influence of the alkyl chain length of ILs on liquid-liquid equilibrium [43–46]. The solubility

of alcohol in ILs increased as the length of the alkyl chain of the IL cation decreased. [MIM][HSO₄] had no alkyl chain on the 3-position of the imidazole ring, and [BMIM][HSO₄] had a butyl group on the 3 position of the imidazole ring, which caused the solubility of methanol in [MIM][HSO₄] to be better than in [BMIM][HSO₄].

From the values of β and S at each temperature for the studied systems, the β values of [BMIM][HSO₄] were greater than those of [MIM][HSO₄], and the S values of [MIM][HSO₄] were greater than those of [BMIM][HSO₄]. Therefore, when a high product purity is required, [BMIM][HSO₄] is a better choice than [MIM][HSO₄]; when large-scale product production is required, [MIM][HSO₄] is a better choice than [BMIM][HSO₄]. Many researchers studied the systems of the MTBE + methanol for the liquid-liquid equilibrium. Urszula et al. and Kazuhiro et al. researched the extraction for separating MTBE and methanol using organic solvents {Boltorn U3000 (B-U3000) and 2,2,4-trimethylpentane (TMP)}, respectively [47–48]. The values of β and S were less than 0.72 and 4.2, respectively. Sang et al. investigated different ILs {1-butyl-3-methylimidazolium tetrafluoroborate ([Bmim][BF₄]) and 1-ethyl-3-methylimidazolium ethyl sulfate (EMISE)} for separating the mixture of MTBE + methanol [49]. The S values of the literature and the system studied in this work were compared. When the concentration of methanol was less than 0.1, [BMIM][HSO₄] and [MIM][HSO₄] had higher β and S than [Bmim][BF₄] and EMISE. When the concentration of methanol was greater than 0.1, the order of S values was EMISE > [MIM][HSO₄] > [BMIM][HSO₄] > [Bmim][BF₄].

3.2. LLE correlation

The NRTL and UNIQUAC models were applied to calculate the correlating parameters of the experimental data for the two systems of MTBE + methanol + [BMIM][HSO₄] and MTBE + methanol + [MIM][HSO₄]. The mole fraction and the activity coefficient (λ) of each component in two phases can be obtained according to Eq. (3).

$$\lambda_i^I x_i^I = \lambda_i^{II} x_i^{II} \quad (3)$$

where x_i^I and x_i^{II} present the mole fraction of component i in the MTBE-rich phase and IL-rich phase, respectively. λ_i^I and λ_i^{II} is the activity coefficients of component i in the MTBE-rich phase and IL-rich phase, respectively. In the NRTL model, the value of nonrandomness factor a_{ij} was fixed to 0.4 and 0.3 for the systems containing [BMIM][HSO₄] and the systems containing [MIM][HSO₄], respectively. In the UNIQUAC model, the structure parameters (volume, r , and surface, q) were taken from the literature [50–52] and are listed in Table 3. The root-mean-square deviations, $rmsd$, are taken as a measurement of the correlation precision. The $rmsd$ equations and the objective function (OF) are expressed as:

$$OF = \sum_{k=1}^M \sum_{j=1}^2 \sum_{i=1}^3 \left(x_{ijk}^{exp} - x_{ijk}^{calc} \right)^2 \quad (4)$$

$$rmsd = 100 \left(\sum_{k=1}^M \sum_{j=1}^2 \sum_{i=1}^3 \frac{\left(x_{ijk}^{exp} - x_{ijk}^{calc} \right)^2}{6M} \right)^{1/2} \quad (5)$$

where M is the number of tie-lines. The x_{ijk}^{exp} and x_{ijk}^{calc} represent the mole fraction of the experiment and calculation, respectively. Subscripts i , j , and k represent the component, phase, and tie-line, respectively. The calculated deviations and the binary interaction parameters for both models are given in Table 4. Correspondingly, the tie-line data calculated by the NRTL and UNIQUAC models are given in Figs. 2 and 3. The $rmsd$ value of NRTL (<0.0131) is lower

Table 2

Experimental LLE data on mass fraction of feed composition w_{IF} , mass fraction w_i , distribution coefficient β and separation factor S for ternary systems of MTBE (1) + methanol (2) + IL (3) at $T = 278.15$ K, 298.15 K^a or 318.15 K under $p = 101.325$ kPa.

Feed (global composition)		Upper phase		Lower phase		β	S
w_{1F}	w_{2F}	w_1	w_2	w_1	w_2		
MTBE (1) + methanol (2) + [BMIM][HSO ₄] (3) 278.15 K							
0.7185	0.0188	0.9758	0.0240	0.0106	0.0435	1.8108	166.535
0.6989	0.0473	0.9549	0.0449	0.0200	0.0806	1.7953	85.8631
0.6843	0.0671	0.9382	0.0613	0.0292	0.1085	1.7691	56.8600
0.6512	0.1106	0.8915	0.1052	0.0436	0.1809	1.7201	35.2081
0.5838	0.1488	0.8462	0.1456	0.0519	0.2121	1.4563	23.7245
0.5176	0.2154	0.7900	0.1977	0.0857	0.2716	1.3739	12.6687
0.4606	0.2926	0.6832	0.2692	0.1385	0.3339	1.2404	6.1184
0.4136	0.3400	0.5655	0.3217	0.2133	0.3776	1.1736	3.1120
MTBE (1) + methanol (2) + [BMIM][HSO ₄] (3) 298.15 K							
0.7196	0.0182	0.9708	0.0185	0.0095	0.0293	1.5854	162.5518
0.6981	0.0477	0.9523	0.0437	0.0188	0.0766	1.7511	88.5760
0.6671	0.0898	0.9231	0.0729	0.0290	0.1345	1.8467	58.7727
0.5841	0.1488	0.8726	0.1211	0.0534	0.2042	1.6869	27.5776
0.5178	0.2153	0.8111	0.1731	0.0922	0.2712	1.5666	13.7810
0.4608	0.2727	0.7424	0.2172	0.1339	0.3178	1.4636	8.1133
0.4144	0.3183	0.5997	0.2933	0.2289	0.3529	1.2032	3.1520
MTBE (1) + methanol (2) + [BMIM][HSO ₄] (3) 318.15 K							
0.7197	0.0187	0.9855	0.0126	0.0093	0.0386	3.0709	325.6579
0.7007	0.0439	0.9633	0.0348	0.0168	0.1063	3.0572	174.8202
0.6667	0.0897	0.9349	0.0618	0.0298	0.1484	2.4002	75.2616
0.5842	0.1482	0.8826	0.1100	0.0615	0.2089	1.8996	27.2626
0.5184	0.2141	0.7656	0.1806	0.1278	0.2783	1.5407	9.2273
0.4590	0.2766	0.6540	0.2530	0.2053	0.3426	1.3540	4.3142
MTBE (1) + methanol (2) + [MIM][HSO ₄] (3) 278.15 K							
0.6220	0.1105	0.8673	0.1327	0.0049	0.0981	0.7393	131.7758
0.5844	0.1488	0.8221	0.1779	0.0051	0.1217	0.6839	111.0725
0.5500	0.1831	0.7821	0.2170	0.0086	0.1482	0.6827	62.2373
0.5171	0.2357	0.7273	0.2711	0.0107	0.1697	0.6261	42.5869
0.4614	0.2922	0.6508	0.3442	0.0176	0.2126	0.6177	22.7929
0.4120	0.3214	0.6324	0.3625	0.0192	0.2233	0.6158	20.2954
0.3686	0.3645	0.5445	0.4383	0.0245	0.2555	0.5830	12.9750
0.3310	0.4025	0.4839	0.4888	0.0268	0.2820	0.5770	10.4052
MTBE (1) + methanol (2) + [MIM][HSO ₄] (3) 298.15 K							
0.6837	0.0671	0.9305	0.0684	0.0050	0.0596	0.8724	163.5747
0.6206	0.1313	0.8485	0.1495	0.0069	0.1093	0.7311	89.5857
0.5387	0.1942	0.7860	0.2079	0.0079	0.1385	0.6660	66.6156
0.4912	0.2515	0.6983	0.2969	0.0088	0.1843	0.6208	49.0258
0.4497	0.3129	0.6262	0.3531	0.0157	0.2186	0.6190	24.7631
0.4120	0.3406	0.5786	0.3995	0.0176	0.2426	0.6073	19.9869
MTBE (1) + methanol (2) + [MIM][HSO ₄] (3) 318.15 K							
0.6220	0.1115	0.8942	0.1050	0.0037	0.1040	0.9906	239.8598
0.5848	0.1488	0.8400	0.1577	0.0052	0.1378	0.8738	140.0897
0.5492	0.1830	0.8009	0.1954	0.0054	0.1644	0.8415	123.7708
0.5179	0.2158	0.7575	0.2348	0.0082	0.1951	0.8312	76.6533
0.4603	0.2619	0.7065	0.2793	0.0091	0.2189	0.7836	61.1433
0.4107	0.3215	0.6156	0.3585	0.0169	0.2631	0.7338	26.6766
0.3679	0.3650	0.5469	0.4074	0.0199	0.3031	0.7440	20.3997

Standard uncertainties u are $u(w) = 0.0119$, $u(T) = 0.05$ K, $u(p) = 1.5$ kPa, $u(m_i) = 0.0006$ g $\delta = \sqrt{\frac{1}{N(N-1)} \sum_{i=1}^N (w_i - \mu)^2}$.

^b From reference [42].

^a From reference [41].

than the *rmsd* of UNIQUAC (<0.0175). Good agreement between the experimental and calculated values are obtained for both models, and relatively larger deviations are observed for a single point for the UNIQUAC model.

4. Conclusions

In this work, LLE for two ternary systems (MTBE + methanol + [BMIM][HSO₄]) and (MTBE + methanol + [MIM][HSO₄]) was determined at 278.15 K, 298.15 K and 318.15 K at 101.325 kPa. The separation efficiency was evaluated by β and S . The values of S for all the systems were more than 1, indicating that the extraction of methanol by the selected ILs from MTBE is feasible. The effect of different temperatures on the extraction capacity of the

two ILs was explored, and the results demonstrate that the higher temperature is more suitable for the extraction. At each temperature, the β value of the MTBE + methanol + [BMIM][HSO₄] system was greater than that for the MTBE + methanol + [MIM][HSO₄] system, and S of the MTBE + methanol + [MIM][HSO₄] system was greater than that of the MTBE + methanol + [BMIM][HSO₄] system. Therefore, [BMIM][HSO₄] was a better choice than [MIM][HSO₄] when a high product purity was required. [MIM][HSO₄] was a better choice than [BMIM][HSO₄] when the production requirements for the product were large. Moreover, comparisons of the *rmsd* values between the data calculated by the NRTL and the UNIQUAC models and the experimental data were performed. The results show that the two models are appropriate for modeling the LLE for the two systems.

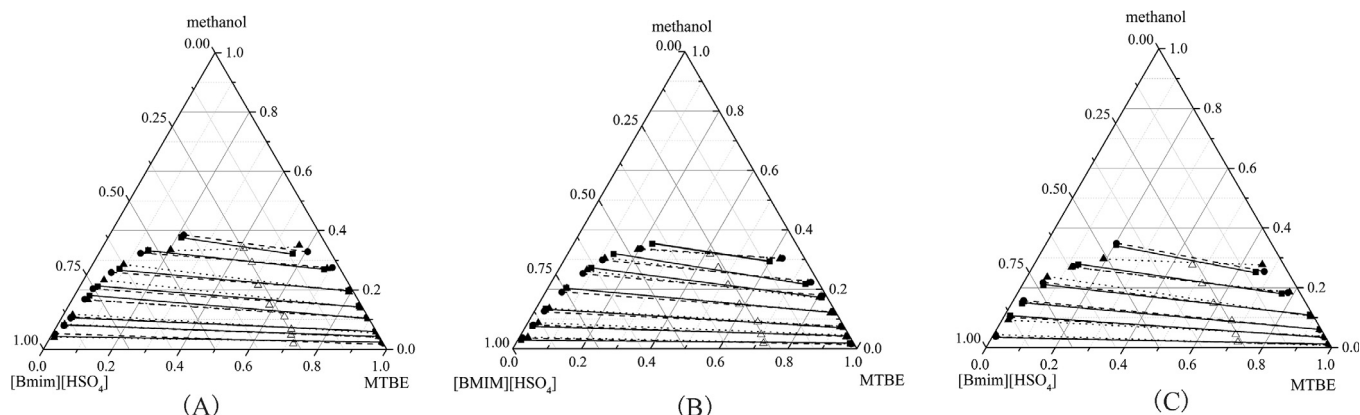


Fig. 2. Experimental and calculated LLE data in mass fraction for the system of MTBE (1) + methanol (2) + [BMIM][HSO₄] (3) (A) at 278.15 K; (■), experimental value; (●), calculated value by NRTL model; (▲), calculated value by UNIQUAC model; (△), mass fraction of feed composition w_{if} ; (B) at 298.15 K; (■), experimental value; (●), calculated value by NRTL model; (▲), calculated value by UNIQUAC model; (△), mass fraction of feed composition w_{if} ; (C) at 318.15 K; (■), experimental value; (●), calculated value by NRTL model; (▲), calculated value by UNIQUAC model; (△), mass fraction of feed composition w_{if} .

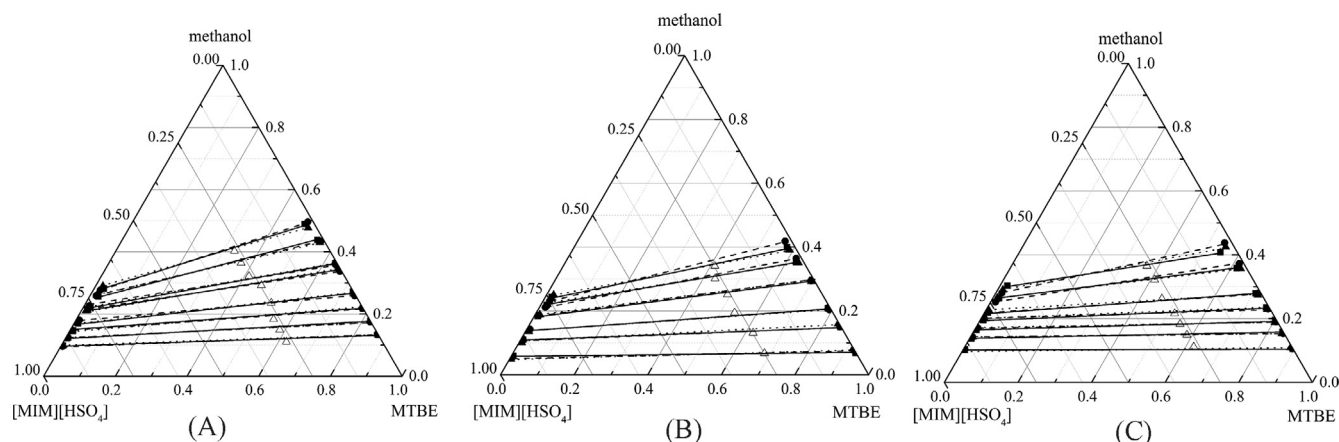


Fig. 3. Experimental and calculated LLE data in mass fraction for the system of MTBE (1) + methanol (2) + [MIM][HSO₄] (3) (A) at 278.15 K; (■), experimental value; (●), calculated value by NRTL model; (▲), calculated value by UNIQUAC model; (△), mass fraction of feed composition w_{if} ; (B) at 298.15 K; (■), experimental value; (●), calculated value by NRTL model; (▲), calculated value by UNIQUAC model; (△), mass fraction of feed composition w_{if} ; (C) at 318.15 K; (■), experimental value; (●), calculated value by NRTL model; (▲), calculated value by UNIQUAC model; (△), mass fraction of feed composition w_{if} .

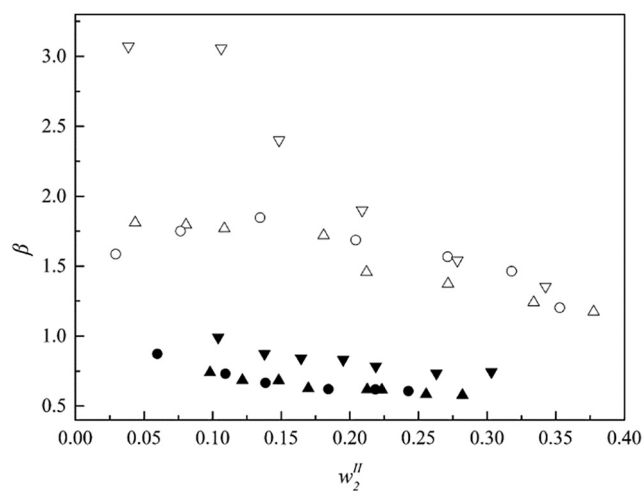


Fig. 4. Distribution coefficient (β) for ternary systems of MTBE + methanol + [BMIM][HSO₄], w_{il} 2 is the mass fraction of methanol in [BMIM][HSO₄]-rich phase. (△), MTBE + methanol + [BMIM][HSO₄] at 278.15 K; (○), MTBE + methanol + [BMIM][HSO₄] at 298.15 K; (▽), MTBE + methanol + [BMIM][HSO₄] at 318.15 K; (▲), MTBE + methanol + [MIM][HSO₄] at 278.15 K; (●), MTBE + methanol + [MIM][HSO₄] at 298.15 K; (▼), MTBE + methanol + [MIM][HSO₄] at 318.15 K.

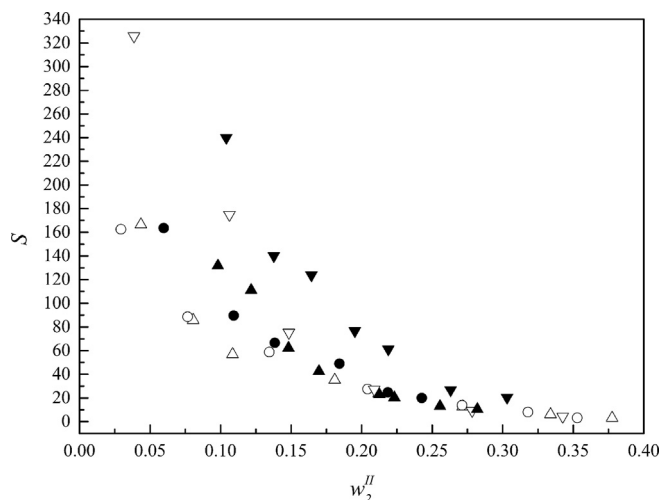


Fig. 5. Separation factor (S) for ternary systems of MTBE + methanol + [BMIM][HSO₄], w_{il} 2 is the mass fraction of methanol in [BMIM][HSO₄]-rich phase. (△), MTBE + methanol + [BMIM][HSO₄] at 278.15 K; (○), MTBE + methanol + [BMIM][HSO₄] at 298.15 K; (▽), MTBE + methanol + [BMIM][HSO₄] at 318.15 K; (▲), MTBE + methanol + [MIM][HSO₄] at 278.15 K; (●), MTBE + methanol + [MIM][HSO₄] at 298.15 K; (▼), MTBE + methanol + [MIM][HSO₄] at 318.15 K.

Table 3Volume (r_i) and surface area (q_i , q_i^r) of UNIQUAC equation.

Component	r^i	q^i	q^r
MTBE ^a	4.562	3.807	3.807
methanol ^a	1.431	1.432	1.432
[MIM][HSO ₄] ^b	5.529	4.505	4.505
[BMIM][HSO ₄] ^c	8.2203	6.628	6.628

^a From reference [50].^b From reference [51].^c From reference [52].**Table 4**NRTL and UNIQUAC models parameters with *rmsd* for ternary mixtures of MTBE, methanol, [BMIM]HSO₄ and [MIM]HSO₄ at 298.15 K and atmospheric pressure.

$i-j$	$\Delta g_{ij}(\text{kJ}\cdot\text{mol}^{-1})$	$\Delta g_{ji}(\text{kJ}\cdot\text{mol}^{-1})$	$rmsd$	α
NRTL parameters				
MTBE (1) + methanol (2) + [BMIM][HSO ₄] (3) at 278.15 K				
1-2	3.7317	7.3400	0.0104	0.40
1-3	23.9722	10.2654		
2-3	11.3336	0.5461		
MTBE (1) + methanol (2) + [BMIM][HSO ₄] (3) at 298.15 K				
1-2	3.8077	5.8144	0.0131	0.40
1-3	14.7203	21.1570		
2-3	10.5170	32.3836		
MTBE (1) + methanol (2) + [BMIM][HSO ₄] (3) at 318.15 K				
1-2	5.5737	7.4991	0.0065	0.40
1-3	19.2694	16.212		
2-3	10.6456	0.5632		
MTBE (1) + methanol (2) + [MIM][HSO ₄] (3) at 278.15 K				
1-2	-2.1405	7.0610	0.0068	0.30
1-3	20.8705	15.9775		
2-3	11.6720	-3.2274		
MTBE (1) + methanol (2) + [MIM][HSO ₄] (3) at 298.15 K				
1-2	2.6872	1.0691	0.0091	0.30
1-3	0.2242	0.4327		
2-3	0.6906	3.1590		
MTBE (1) + methanol (2) + [MIM][HSO ₄] (3) at 318.15 K				
1-2	3.0023	9.1433	0.0091	0.30
1-3	21.5067	15.0465		
2-3	11.5504	1.7369		
$i-j$	$\Delta u_{ij}(\text{kJ}\cdot\text{mol}^{-1})$	$\Delta u_{ji}(\text{kJ}\cdot\text{mol}^{-1})$	$rmsd$	
UNIQUAC parameters				
MTBE (1) + methanol (2) + [BMIM][HSO ₄] (3) at 278.15 K				
1-2	1.2953	-0.2843	0.0121	
1-3	2.9013	1.4368		
2-3	-2.9541	6.7755		
MTBE (1) + methanol (2) + [BMIM][HSO ₄] (3) at 298.15 K				
1-2	3.6710	0.8565	0.0175	
1-3	1.5512	0.8665		
2-3	3.7035	0.1550		
MTBE (1) + methanol (2) + [BMIM][HSO ₄] (3) at 318.15 K				
1-2	3.8662	-1.2791	0.0160	
1-3	1.7746	1.8397		
2-3	-0.7306	-0.3536		
MTBE (1) + methanol (2) + [MIM][HSO ₄] (3) at 278.15 K				
1-2	2.8376	-0.7139	0.0048	
1-3	5.4530	1.4647		
2-3	0.8561	0.3695		
MTBE (1) + methanol (2) + [MIM][HSO ₄] (3) at 298.15 K				
1-2	2.0317	-0.9664	0.0134	
1-3	3.2836	2.0297		
2-3	4.8676	-1.9014		
MTBE (1) + methanol (2) + [MIM][HSO ₄] (3) at 318.15 K				
1-2	2.3453	1.2497	0.0080	
1-3	2.2661	2.4037		
2-3	2.0870	0.9067		

^a Standard uncertainties u are $u(T) = 0.05$ K.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jct.2018.12.029>.

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